

Original Article

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Experimental studies on the VLE of R-32/R-125 and verification of experimental setup

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Abstract Alternative refrigerants for R410A, which is most commonly used in building cooling and heating, are being requested with a global warming potential of 750 or lower and these regulations will continue to be implemented. At the current stage, several candidates are being discussed, but these will not be the final solution. What is certain is the R410A replacement refrigerant will be proposed in the form of mixed refrigerants. To figure out the thermodynamic properties of the mixed refrigerant, VLE data must be obtained first. In this study, we performed VLE experiments on a representative mixed refrigerant (R32-R125) to verify our newly constructed VLE experiment facility, and compared the results with the REFPROP 10 and the results of other researchers. The experiment was conducted at a temperature of 273.15 K to 313.15 K, and the maximum MAPE for pressure was found to be 0.0178 % and the maximum MAE for composition was found to be 0.0029. In the future, the facility will be used to present VLE test data with new mixed refrigerants as candidate replacement refrigerants for R410A.

1. Introduction

Since 2010, when HFO-based refrigerants with low GWP (global warming potential) became commercially available, there has been a significant change in the HVAC & R market, and chiller (turbo, screw) products targeting low GWP refrigerants have been commercialized by several manufacturers. In the chiller market using low-pressure refrigerant (R123) and medium-pressure refrigerant (R134a), final candidates have been determined, but the alternative refrigerant for R410A is not yet clear. The GWP value of alternative refrigerants for R410A, which is most commonly used in the cooling and heating of buildings is required to be less than 750, and there is a recent movement in the EU to require this to be less than 150. Several candidates are being discussed, including R32, R454B, and R466A, but these are also likely to be mid-term solutions. However, what is certain is that the direction of the R410A replacement refrigerant will be presented in the form of a mixed refrigerant, and to derive the thermodynamic characteristics of the mixed refrigerant, VLE data must be obtained first.

In Korea, in the early 2000s, due to interest in natural refrigerant, there was VLE research on mixed refrigerants using propane and CO₂ as raw materials [1, 2], but unfortunately, follow-up VLE research targeting HFO series is not underway. However, overseas, VLE research targeting HFO-based refrigerants has been conducted since 2010, when HFO refrigerants were introduced, and VLE research using R1234yf or R1234ze(E) as a raw material [3–8] was conducted the most. Recently, VLE research [9–15] using R1123 and R1311 as raw materials is also being continuously studied.

In this study, we performed a VLE experiment on a representative mixed refrigerant (R32-R125) to verify our newly built VLE experiment facility and compared the results with the REFPROP 10 and those of other researchers.

2. Experimental apparatus and procedures

2.1 VLE experimental apparatus

In order to accurately predict the thermodynamic properties of a mixed refrigerant, the binary interaction parameters between the two pure refrigerants, which are the components mixture, must be known. The VLE experiment is required to develop the equation of state for mixed refrigerants, from which the temperature, pressure and composition of liquid and gas are measured in an equilibrium state. Fig. 1 shows a schematic diagram of the VLE experiment facility, and Table 1 shows the specifications of the main equipment. VLE experimental equipment largely consists of an equilibrium cell, temperature sensor, pressure sensor, constant temperature bath, and gas chromatography for composition measurement. The equilibrium cell has an internal volume of approximately 150 cm³ and is equipped with a window to check the gas-liquid interface. The refrigerant temperature inside the equilibrium cell and the external fluid temperatures were measured with a precision platinum resistance thermometer, respectively (secondary reference PRT) and the refrigerant pressure inside the equilibrium cell was measured with a precision pressure gauge

(0.01 % of F.S). Special care was required when measuring pressure, and the pressure gauge was submerged inside the bath to prevent local condensation and evaporation that may occur through heat transfer with ambient air. All temperature and pressure sensors were connected to a readout device that was connected to a computer through a serial communication to record digital values. A constant temperature bath was set based on the measured temperature of the equilibrium cell, and was maintained within approximately ± 0.01 K based on the set temperature. The expanded uncertainties of pressure and temperatures were ± 0.9 kPa and ± 0.03 K.

2.2 Peak area calibration

Measuring composition in gas chromatography requires professional care and is the most sensitive part of the VLE experiment. The column in the GC was composed of PoraPLOT Q (CP7551, capillary), and was detected through a thermal conductivity detector (TCD). Carrier gas was He gas (99.9999 %), and split ratio was set to 100. Generally, the area fraction analyzed in gas chromatography has a strong depend-

Table 1. Specification of main equipment.

Item	Model	Specification
Platinum resistance thermometer	Fluke 5608 (bath)	Spec: -200 to 500 °C Accuracy: ± 0.008 K (@0 °C)
	Fluke 5609 (refrigerant)	
Thermometer readout	Fluke 1529	Spec: 0 to 400 Ω Accuracy: ± 25 ppm of reading
Pressure transmitter	Paroscientific, 31K-101	Spec: 0 to 6.9 MPa Accuracy: ± 0.01 % of span
Pressure readout	Paroscientific, model 735	Accuracy: ± 0.0001 % of span
Constant temperature bath	Kambic OB-50/2 LT	Spec: -40 to 130 °C Stability: ± 0.005 K (@20 °C)
Gas chromatography	Agilent 8890 GC	Column: PoraPlot Q (capillary) Detector: TCD

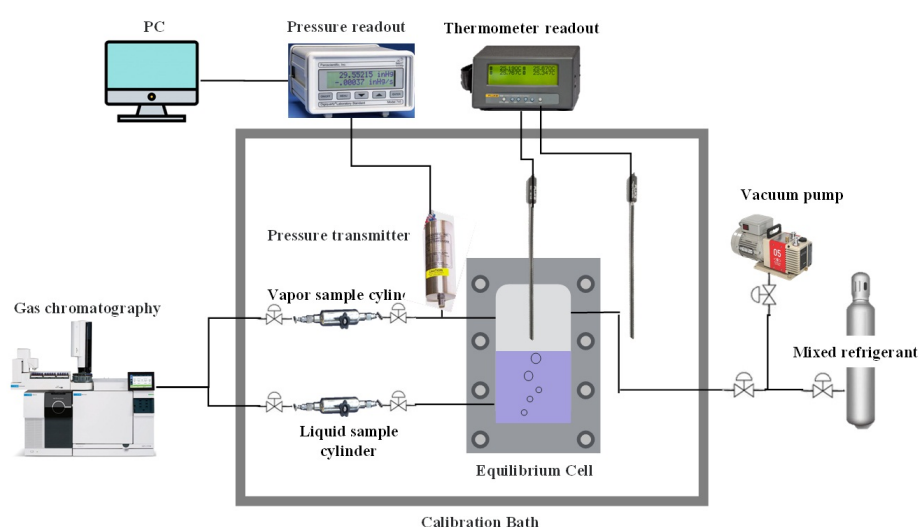


Fig. 1. Schematic diagram of VLE apparatus.

ency on the mole fraction, but since it does not react linearly, the area fraction must be calibrated. A sample mixed refrigerant was prepared using the gravimetric method, and the refrigerant used here is a pretreated so that it becomes of high-purity pure refrigerant. After creating a vacuum inside the sample cylinder, refrigerants with low saturation pressures were injected in that order, R125 and R32, and the weight before and after injection was measured using a precision scale. The area fractions were measured more than ten times for mixed refrigerant samples with more than five different compositions, and a third-order polynomial fit was produced based on the results. Ultimately, the expanded uncertainty in the composition (mole fraction) of the mixed refrigerant through area calibration was confirmed to be within ± 0.0013 .

2.3 Experimental procedures and conditions

The purity of refrigerants commercially available in Korea is over 99.5 % by domestic standards. In order to measure exact refrigerant properties, impurities must be removed to further increase purity. In general, the freeze-pump-thaw method is used to increase the purity of the refrigerant, and high-purity pure refrigerant was manufactured by referring to the gas preparation procedure [16]. Freeze procedure means immersing the cylinder in a liquid nitrogen container to solidify the refrigerant. Afterwards, the vacuum pump was operated to remove non-condensable gases (pump), and the solidified refrigerant was melted back to room temperature (thaw) to separate impurities dissolved in the refrigerant. The cycle was repeated more than two times, and the purity of the final refrigerant was confirmed to be more than 99.99 % through gas chromatography.

After maintaining the vacuum inside the equilibrium cell, the mixed refrigerant prepared through the gravimetric method was injected, and the appropriate amount was injected while checking the vapor-liquid interface. When thermal equilibrium was reached, the temperature and pressure of the refrigerant were measured, and then some samples of the liquid and gas in the equilibrium cell were extracted into the sample cylinder.

The pressure was measured during the extraction process and the extraction amount was adjusted so that the mixed refrigerant sample was always maintained in a gaseous state in the sample cylinder. The composition of the mixed refrigerant extracted from the sample cylinder was analyzed at least five times, and the experiment was repeated with the same procedure by adjusting the temperature of the bath.

Vapor-liquid equilibrium data were measured in the range of 273.15 K to 313.15 K for a total of 6 samples, including four mixed refrigerants with different compositions and two pure refrigerants.

3. Experimental results and discussion

3.1 VLE experimental results

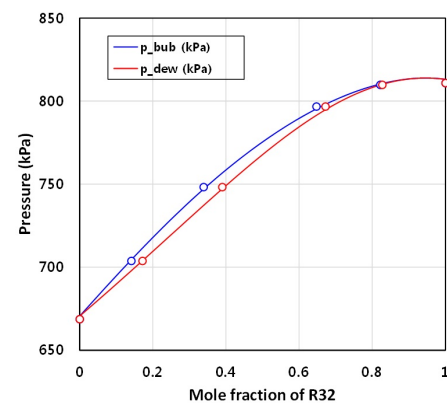
Experiments were performed at 10 K intervals in the saturation temperature range of 273.15 to 313.15 K for samples with different mixed refrigerant compositions (mole fractions). Table 2 shows the temperature, pressure, and composition values of the mixed refrigerant, where x means liquid, y means gas composition, and subscript 1 means R32 with high vapor pressure. Analysis of deviations (pressure, vapor mole fraction) between experimental values and REFPROP 10 [17] was performed as follows. For the deviation analysis the value presented generally refers to the largest value among MAE (composition of y) and MAPE (pressure) calculated at each temperature [12]. The procedure for calculating the deviation is as follows.

1) Calculate the bubble point pressure ($P_{cal,i}$) in the REFPROP 10 using the measured $x_{1,exp}$ and T_{exp} , and calculate the deviation from the measured pressure ($P_{exp,i}$)

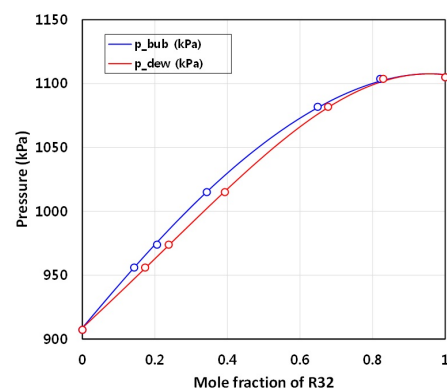
$$\text{MAPE}(P) = \frac{100}{N} \sum \left| \frac{P_{exp,i} - P_{cal,i}}{P_{exp,i}} \right| (\%). \quad (1)$$

2) Using the $P_{cal,i}$, the $y_{1,cal}$ can be obtained in REFPROP 10 and the deviation from the measured $y_{1,exp}$ is calculated.

$$\text{MAE}(y) = \frac{1}{N} \sum |y_{exp,i} - y_{cal,i}| (-). \quad (2)$$



(a) P-x diagram at $T = 273.15$ K



(b) P-x diagram at $T = 283.15$ K

Fig. 2. P-x diagram for R32/R125 mixed refrigerant.

Table 2. VLE data measured for R32 and R125 mixture and comparison with REFPROP 10.

T_{exp} (K)	$x_{1,exp}$ (-)	$y_{1,exp}$ (-)	P_{exp} (kPa)	P_{cal} (kPa)	$y_{1,cal}$ (-)	dy_1 (-)	dP (%)	MAE y_1 (-)	MAPE P (%)
273.15	0	0	668.67	670.46	0	0	-0.2673	0.0027	0.1779
273.15	0.141	0.172	703.75	704.45	0.175	-0.0027	-0.1		
273.15	0.339	0.39	748.25	746.99	0.389	0.0007	0.1687		
273.14	0.647	0.672	796.73	795.03	0.676	-0.0042	0.2145		
273.15	0.822	0.827	809.75	810.31	0.831	-0.0032	-0.0681		
273.14	1	1	810.88	812.9	1	0	-0.2488		
283.16	0	0	907.2	909.01	0	0	-0.1996	0.0027	0.1197
283.16	0.143	0.173	955.91	956.17	0.175	-0.0021	-0.0272		
283.15	0.206	0.238	973.86	975.66	0.247	-0.0088	-0.1841		
283.15	0.343	0.393	1014.97	1015.24	0.391	0.0012	-0.0272		
283.15	0.649	0.677	1081.75	1081.08	0.676	0.0009	0.062		
283.15	0.821	0.829	1103.69	1101.94	0.83	-0.0005	0.1589		
283.15	1	1	1104.9	1106.87	1	0	-0.1786	0.0024	0.0950
293.15	0	0	1203.4	1205.22	0	0	-0.1506		
293.15	0.146	0.175	1269.62	1270.18	0.177	-0.002	-0.0437		
293.15	0.203	0.237	1292.76	1293.99	0.241	-0.0044	-0.0952		
293.15	0.345	0.389	1348.04	1349.2	0.391	-0.0023	-0.0861		
293.15	0.648	0.677	1438.44	1437.51	0.674	0.0023	0.0648		
293.15	0.822	0.829	1468.33	1466.38	0.83	-0.0013	0.1328	0.0011	0.0749
293.18	1	1	1474.25	1475.61	1	0	-0.092		
303.15	0	0	1565.81	1568.47	0	0	-0.1703		
303.15	0.137	0.164	1649.18	1648.85	0.164	-0.0003	0.0201		
303.15	0.19	0.222	1678.47	1678.39	0.224	-0.0015	0.0048		
303.15	0.335	0.374	1752.05	1753.29	0.377	-0.0027	-0.0711		
303.15	0.643	0.669	1875.56	1874.28	0.668	0.0008	0.0682	0.0029	0.1142
303.15	0.818	0.827	1917.36	1914.58	0.827	0.0001	0.145		
303.16	1	1	1927	1927.86	1	0	-0.0447		
313.15	0	0	2000.73	2008.45	0	0	-0.3859		
313.15	0.186	0.211	2148.24	2147.4	0.216	-0.0052	0.0392		
313.15	0.64	0.664	2406.06	2404.61	0.663	0.0006	0.0603		
313.15	0.816	0.827	2460.38	2458.67	0.824	0.0029	0.0693	0.0029	0.1142
313.15	1	1	2477.96	2478.37	1	0	-0.0163		

Specifically, it was calculated using the REFPROP function provided by Excel, and can be calculated as $P_{cal} = \text{Pressure}(x, \text{"Tliq"}, T_{exp})$, $y_1 = \text{VaporMoleFraction}(x, \text{"Tliq"}, T_{exp})$.

Fig. 2 compares some of the experimental data values (273.15 K, 283.15 K) with the results of REFPROP 10, and the $P-x$ diagram indicates the change in bubble pressure and dew pressure depending on the composition in the isotherm state. The bubble pressure and dew pressure calculated with REFPROP 10 are drawn as blue and red solid lines, and the circle mark represents the experimental results.

Overall, the experimental results can be confirmed to be in good agreement with the calculation results of the REFPROP 10, and as shown in Table 2, the maximum MAPE (pressure) was confirmed to be 0.0178 % and the maximum MAE (composition) was confirmed to be 0.0029.

3.2 Comparison to other reference

Using the VLE data referenced in the equation of state study of R32-R125 mixture, a deviation analysis was performed between the data and the REFPROP 10 using the same method. Lemmon [18] introduced literatures referenced in the REFPROP 10, and in the case of R32-R125, eight VLE data sets were referred to. Holcomb et al. [19] provides a total of 30 VLE data under five isotherm conditions in the range of 279 K to 341 K, and Kato. et al. [20] provides a total of 39 VLE data under six isotherm conditions in the range of 318.15 to 349.15 K.

Fig. 3 shows the deviation of pressure and composition according to the mixed refrigerant composition, and Fig. 4 shows the deviation of pressure and composition according to saturation.

Table 3. Comparison summarizations with previous literatures.

Item	Data points	Temperature (K)	Maximum MAPE (P, %)	Maximum MAE (y_1 , -)
This study	32	273.15-313.15	0.1779	0.0029
Holcomb et al. [19]	30	280.29-340.08	0.3284	0.0048
Kato et al. [20]	39	318.15-349.15	1.0776	0.0034

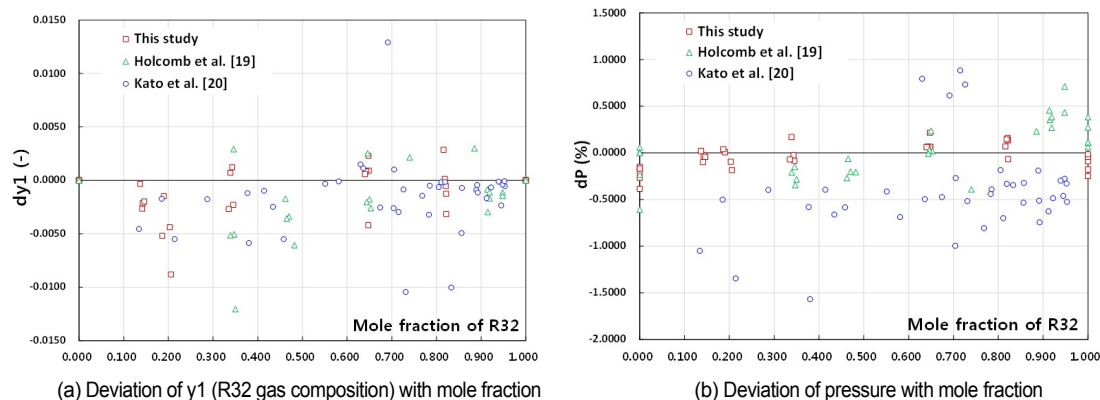


Fig. 3. Deviation of composition and pressure with previous studies (with compositions).

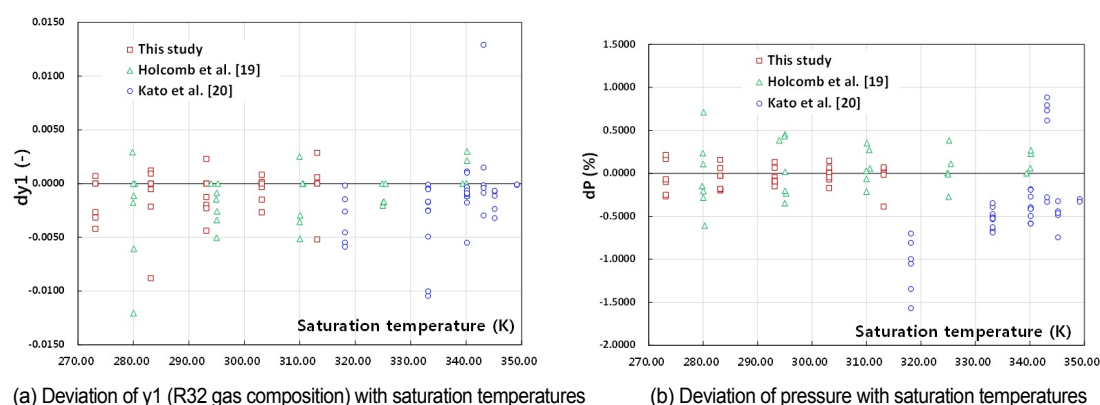


Fig. 4. Deviation of composition and pressure with previous studies (with saturation temperature).

tion temperature. Table 3 summarizes the maximum composition MAE and pressure MAPE values. It was confirmed that the experimental results presented in this study show an accuracy level equal to or higher than that of previous studies referenced in REFPROP 10, and based on this, the validity of the experimental measurement and analysis could be confirmed.

4. Conclusions

Alternative refrigerants for R410A, which is most commonly used in building cooling and heating, are being requested with a global warming potential of 750 or lower and these regulations will continue to be implemented. At the current stage, several candidates are being discussed, but these will not be the final solution, and what is certain is the R410A replacement refrigerant will be proposed in the form of mixed refrigerants, and for this purpose, we have built a test facility to measure

VLE data of mixed refrigerants. In this study, we performed a VLE experiments on a representative mixed refrigerant (R32-R125) to verify our newly built experimental facility, and compared the results with the REFPROP 10 and the results of other researchers. The performance test was conducted at a temperature of 273.15 K to 313.15 K, and the MAPE for pressure was confirmed to be 0.0178 % and the MAE for composition was confirmed to be 0.0029. It was confirmed that the experimental results presented in this study show an accuracy level equal to or higher than that of previous studies referenced in REFPROP 10. In the future, the facility will be used to present VLE experiment data with a new mixed refrigerant as an alternative refrigerant of R410A.

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Nomenclature

P	: Pressure
T	: Temperature or Celsius temperature scale
x	: Composition of liquid in mixed refrigerant
y	: Composition of vapor in mixed refrigerant

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